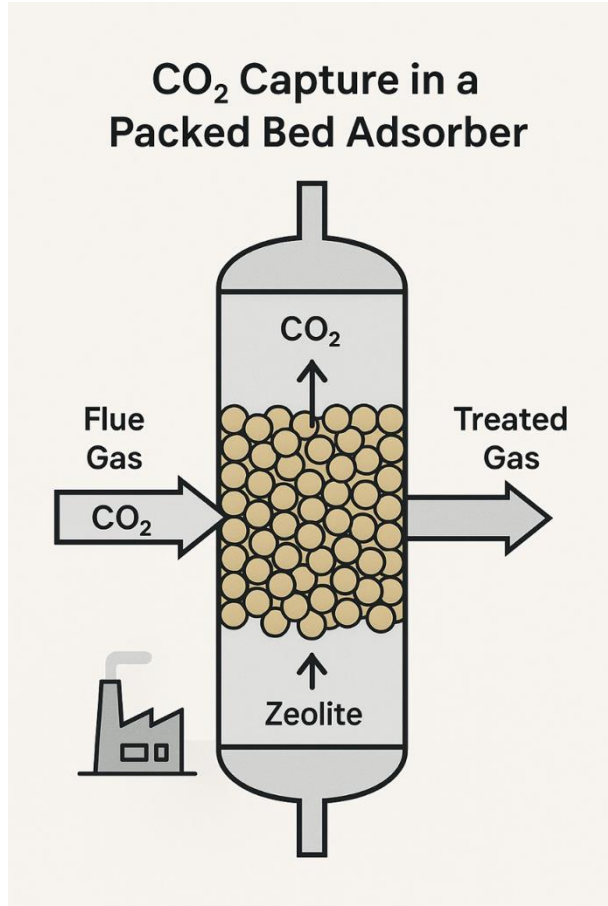




CO₂ ADSORPTION SIMULATION IN PACKED BED COLUMNS

Modeling and Visualizing CO₂ Capture
Technologies

Problem Statement: Climate Change and CO₂ Emissions

- ▶ The increasing levels of CO₂ emissions contribute significantly to global warming, prompting urgent measures for reduction. In industries, post-combustion processes are major sources of CO₂, necessitating innovative capture technologies.

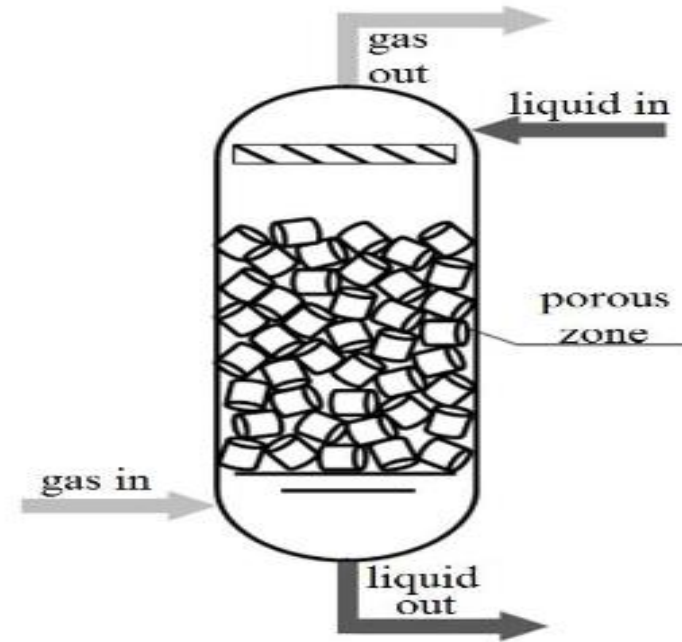
Industrial Context: Post-combustion CO₂ Capture

- Post-combustion CO₂ capture technologies, such as packed bed adsorbers, effectively target flue gases emitted from power plants. This method selectively removes CO₂, presenting an opportunity for cleaner industrial processes.

INDUSTRIAL CONTEXT-

- ▶ *Power plants relying on fossil fuels emit large amount of CO₂ part of **FLUE GAS**- generally includes CO₂ , N₂, H₂O, O₂, Trace pollutants*
- ▶ *CO₂ is a major contributor to global warming, capturing it before it enters the atmosphere is a key part of climate change mitigation strategies*
- ▶ *One promising technology for **post-combustion carbon capture-Adsorption-Based Systems**, CO₂ is selectively removed from the flue gas using solid adsorbents such as Zeolites*
- ▶ **Zeolites-**
- ▶ microporous, crystalline aluminosilicates with high surface area and strong affinity for CO₂

WHAT IS THE PROCESS?



(Zeolites are shaped as pellets in Packed- Bed Columns)

As CO₂ flows -

1. CO₂ adsorbed on surface of Zeolite Pellets, reducing their concentration in gas stream
2. Other components pass through with minimal adsorption.

3. Adsorption **not** a **static process**- it changes over time

4. Time dependent behavior termed as **Transient Adsorption**

5. This is what our model is about-

- . **Designing efficient capture systems**
- . **Predicting breakthrough curves** (when CO₂ starts to appear in the outlet gas)
- . **Optimizing regeneration cycles** (when and how to refresh the adsorbent)

#. Breakthrough Point: When CO₂ first appears at the outlet of the packed bed – the bed starts to lose efficiency.

#. Mass Transfer Zone (MTZ): The active region in the bed where CO₂ is being adsorbed. It moves as the bed saturates.

GOVERNING EQUATIONS

1. Mass Balance (PDE - Convection-Diffusion-Reaction):

$$\epsilon \frac{\partial C}{\partial t} = D_{ax} \frac{\partial^2 C}{\partial z^2} - u \frac{\partial C}{\partial z} - (1 - \epsilon) \rho_s \frac{\partial q}{\partial t}$$

- $C(z,t)$: CO₂ concentration in gas phase (mol/m³).
- $q(z,t)$: Adsorbed CO₂ on zeolite (mol/kg).
- D_{ax} : Axial dispersion coefficient.
- u : Superficial velocity (m/s).
- ϵ : Bed porosity.

This equation tracks how CO₂ gas concentration $C(z,t)$ changes in the packed bed due to:

- Axial dispersion : $D_{ax} \frac{\partial^2 C}{\partial z^2}$
- Convection : $u \frac{\partial C}{\partial z}$
- Adsorption : $(1 - \epsilon) \rho_s \frac{\partial q}{\partial t}$

2. **Adsorption Kinetics (ODE - Langmuir Model):**

$$\frac{dq}{dt} = k_a C(q_{max} - q) - k_d q$$

- k_a, k_d : Adsorption/desorption rate constants.
- q_{max} : Saturation capacity.

This equation describes how CO₂ adsorbs onto the solid:

- **Adsorption term:** $k_a C(q_{max} - q)$
- **Desorption term:** $- k_d q$

How to solve?

1). Spatial Discretization of the PDE-

Dividing the column into N nodes along the spatial domain
z

Approximate the spatial derivatives using **finite differences method (FDE)**

$$\frac{\partial C}{\partial z} \approx \frac{C_{i+1} - C_{i-1}}{2\Delta z}$$

$$\frac{\partial^2 C}{\partial z^2} \approx \frac{C_{i+1} - 2C_i + C_{i-1}}{\Delta z^2}$$

- 2) This discretization turns the PDE into a system of **ODEs** in time for each spatial point.
- 3) For each spatial node i , we have

$$\epsilon \frac{dC_i}{dt} = D_{ax} \frac{C_{i+1} - 2C_i + C_{i-1}}{\Delta z^2} - u \frac{C_{i+1} - C_{i-1}}{2\Delta z} - (1 - \epsilon)Q_s \frac{dq_i}{dt}$$

- 4) The **ODE** for absorption kinetics is used to compute derivative of q_i with respect to t at each node

$$\frac{dq_i}{dt} = k_a C_i (q_{max} - q_i) - k_d q_i$$

Then to solve coupled differential equations, we can use the Explicit Euler Method, Implicit Euler Method, Runge Kutta 4 or ODE15s Method in vector form involving matrices to solve it.

App Interface

If we take Default entries, the app shows the following :

Model Parameters (Optimized)

Column length (m)

1

Spatial points (n)

200

Velocity u (m/s)

0.01

Dispersion D_{ax} (m²/s)

1e-06

Porosity ϵ

0.35

Density ρ_s (kg/m³)

50

Capacity q_{max} (mol/kg)

1

Adsorption k_a (1/s)

0.01

Desorption k_d (1/s)

0.005

Inlet C_0 (mol/m³)

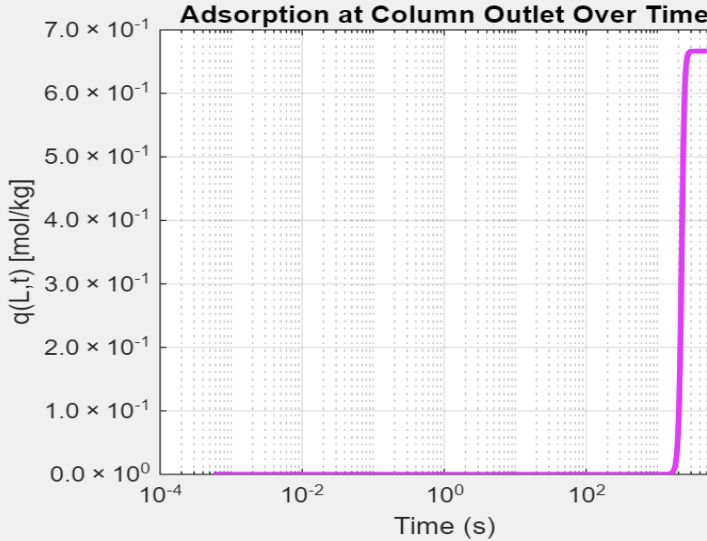
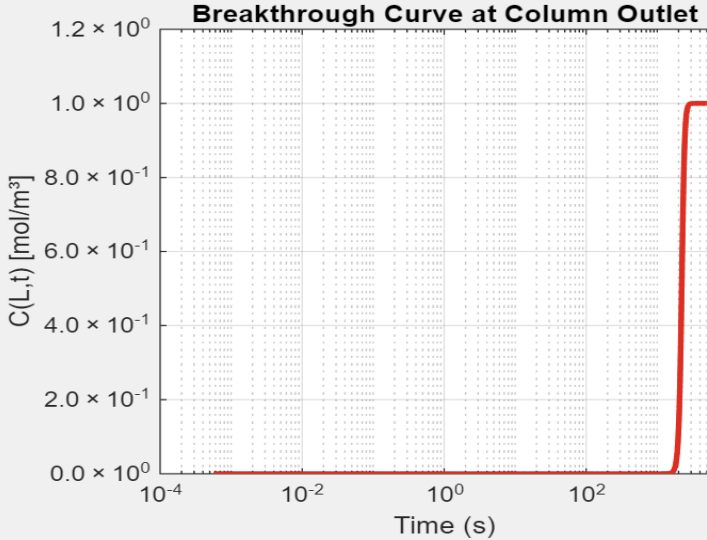
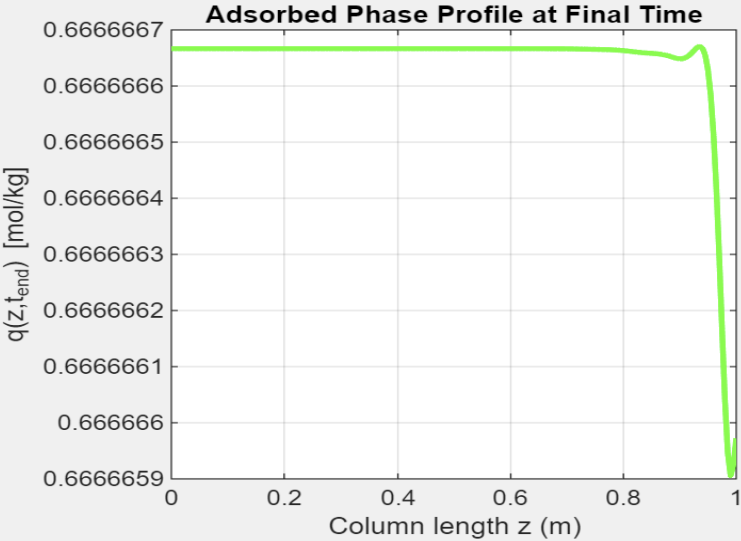
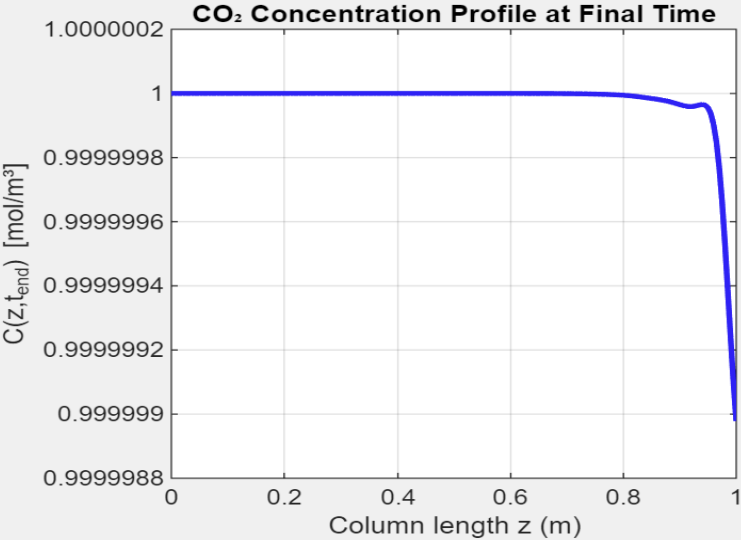
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Time span (s)

5000

Run Simulation

Results



Graph explanation

C vs z graph :

1.Flat region

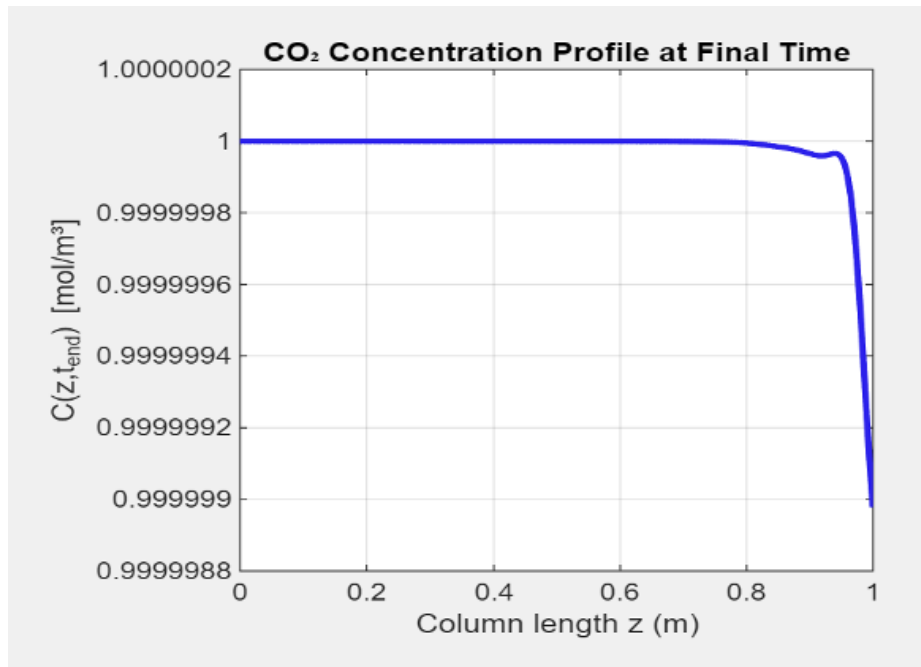
CO₂ concentration remains very close to 1 mol/m³, ~the inlet value, because in initial region c₂ is fully adsorbed .

2.Sharp drop :

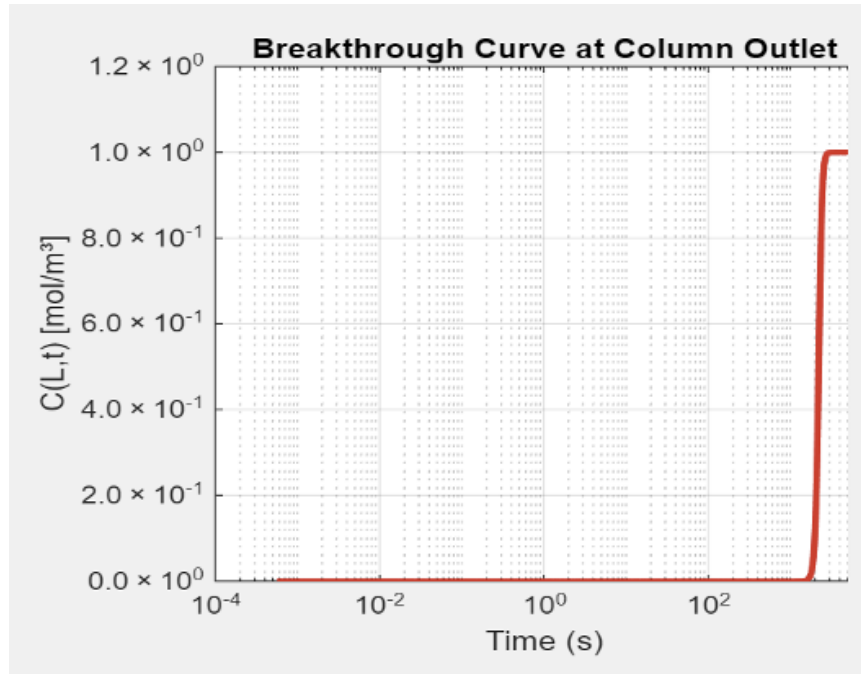
CO₂ concentration suddenly drops — from almost 1 to a much lower value.

This is the "mass transfer zone", where gas has now reached:

Adsorbent is nearing saturation.



C vs time :



1. Flat region ($C \approx 0$):

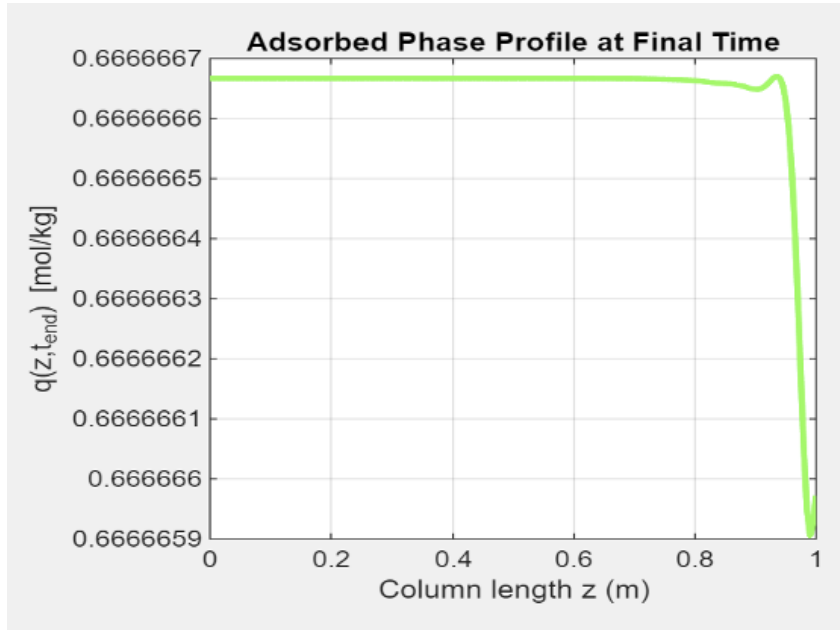
Initially, no CO_2 escapes the column.
Because the adsorbent is fresh and efficient.

2. Sharp rise:

The adsorption front reaches the end of the column.
 CO_2 begins to leak from the outlet.
This is the breakthrough point.

3. Plateau at the top ($C \approx C_0$):

Eventually, the entire bed is saturated.
 CO_2 passes through without being adsorbed.
Outlet concentration equals inlet concentration .



q vs z Graph:

1. left side of graph

This is the inlet.

q is maximum here ($\approx q_{\text{max}}$) $\rightarrow$ the adsorbent is fully loaded.

It has been exposed to CO_2 the longest.

2. Right side of the column ($z \sim 1$):

This is the outlet.

q is low here (maybe even ≈ 0) $\rightarrow$ little or no CO_2 has reached here yet.

So the adsorbent here is still fresh.

q vs t Graph:

1. Initial Time :

$q = 0 \rightarrow$ as No CO_2 has reached the outlet yet.

The gas is being adsorbed upstream (closer to inlet), so the outlet adsorbent is still fresh.

It takes time to reach the end (outlet).

2. Around $t \approx 5000$ s:

Sudden rise in q , as CO_2 finally reaches the outlet.

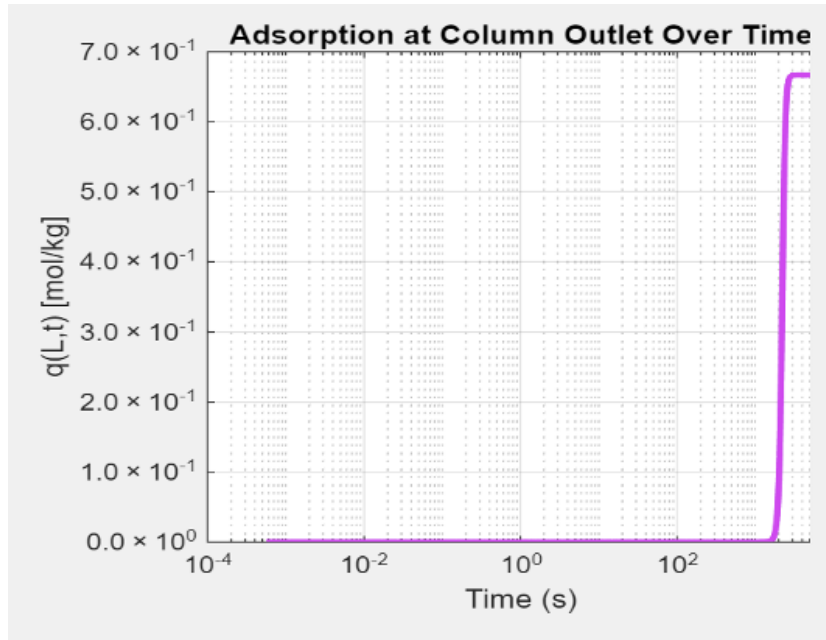
Adsorption begins in the last section of the bed.

The bed can no longer capture all the CO_2 — some starts leaking out from the outlet.

3. Post ~ 5000 s:

q at outlet rises steeply, and then gradually flattens (approaches q_{max}).

Eventually, it saturates and q reaches its maximum capacity.



By -

Haryashva Gupta	(240445)
Nikita Nehra.	(240698)
Harshit Pandey	(240441)
Kritika Singh	(240568)
Arjit Agarwal.	(240180)
Anubhav Jha	(240152)
Ayushi Sharma.	(240247)
Deepanshu Kashyap.	(240332)
Jugal Pahuja.	(240501)
Viraaj Pathariya.	(241169)

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